

Uncertain Volatility: Price-Level vs Log-Moneyness Fitting — Results

1. Setup

Two parameterisations of a 3-state uncertain volatility model were compared on the same ATM 1M IV data across four SPX volatility regimes:

	Log-Moneyness (Displaced Diffusion)	Price-Level (Bachelier Mixture)
States	3	3
Params per state	weight, σ , σ_{ref}	weight, σ_N
Total params	9	6
ATM price	State i: DD call with σ , σ_{ref} , $\sigma_n = \sigma_{ref} \cdot S_{ref}$	State i: Bachelier call with σ_N
Output	Mixture price $\rightarrow$ BS IV	Mixture price $\rightarrow$ BS IV

2. Fitted Parameters

Log-Moneyness Model (9 params)

Regime	w	w	w	σ_{ref}	σ	σ_n	σ_{ref}	σ	σ_n	MSE
Low Vol (2015-17)	0.387	0.340	0.272	0.895	0.901	0.100	0.010	0.010	0.343	0.000795
Transition (2018-19)	0.384	0.338	0.278	0.884	0.891	0.100	0.010	0.010	0.401	0.000907
COVID	0.254	0.261	0.486	0.135	0.100	0.100	0.010	0.010	0.236	0.005632
Post-COVID	0.377	0.333	0.291	0.912	0.872	0.184	0.010	0.010	0.517	0.002463

Price-Level Model (6 params)

Regime	w	w	w	σ_N	σ_N	σ_N	MSE
Low Vol (2015-17)	0.190	0.201	0.609	50.0	150.0	400.0	0.000722

Regime	w	w	w	_N	_N	_N	MSE
Transition (2018-19)	0	0	1.000	50.0	150.0	400.1	0.000904
COVID	0	0	1.000	50.1	150.1	703.3	0.005488
Post-COVID	0	0	1.000	50.1	150.2	776.3	0.002549

3. Effective Normal Volatility Comparison

Effective normal volatility = $_ln \times S_mean$ for log-moneyness; $_N$ directly for price-level. Units: price/ $\sqrt{yr}$.

Regime	Model	_eff,	_eff,	_eff,	MSE
Low Vol (\$ 2198)	Log-moneyness	22.0	22.0	754.7	0.000795
Low Vol	Price-level	50.0	150.0	400.0	0.000722
Transition (\$ 2827)	Log-moneyness	28.3	28.3	1132.9	0.000907
Transition	Price-level	50.0	150.0	400.1	0.000904
COVID (\$ 3239)	Log-moneyness	32.4	32.4	763.5	0.005632
COVID	Price-level	50.1	150.1	703.3	0.005488
Post-COVID (\$ 4637)	Log-moneyness	46.4	46.4	2396.4	0.002463
Post-COVID	Price-level	50.1	150.2	776.3	0.002549

4. MSE Comparison

Regime	n_obs	Log-Moneyness MSE	Price-Level MSE	Δ MSE
Low Vol (2015-17)	705	0.000795	0.000722	−9.2%
Transition (2018-19)	471	0.000907	0.000904	−0.3%
COVID	234	0.005632	0.005488	−2.6%
Post-COVID	995	0.002463	0.002549	+3.5%

5. Conclusions

5.1 The price-level model is degenerate at ATM

For 3 of 4 regimes, the optimiser collapses the mixture weights to (0, 0, 1.0), reducing the 3-state model to a **single-state Bachelier model**. This is a mathematical consequence of the ATM call pricing formula:

$$C_{\text{Bachelier}}(S=K, r, \sigma_N, T) = e^{-rT} \cdot \sigma_N \sqrt{T} \cdot \phi(0)$$

Under a mixture, the ATM price becomes:

$$C_{\text{mix}} = e^{-rT} \sqrt{T} \cdot \phi(0) \cdot \sum_i w_i \sigma_{N,i}$$

This is observationally equivalent to a single state with effective volatility $\sigma_{\text{eff}} = \sum w_i \sigma_{N,i}$. The individual states are not identifiable — the optimiser correctly finds that the simplest representation suffices.

5.2 $_N$ parameters are stuck near initial conditions

Initialised at [50, 150, 400], the fitted values barely move (2–10 iterations):
 - Low Vol: [50.0, 150.0, 400.0] - Transition: [50.0, 150.0, 400.1] -
 COVID: [50.1, 150.1, 703.3] - Post-COVID: [50.1, 150.2, 776.3]

The loss surface is essentially flat in $_N$ space when weights collapse — only the *weighted sum* matters, not the individual components.

5.3 MSE is nearly identical between the two models

Despite the structural difference (9 vs 6 params), both models achieve comparable MSE across all regimes (within $\pm 10\%$). This confirms the theoretical prediction: at ATM, both parameterisations reduce to fitting a single effective volatility curve. The extra parameters in the log-moneyness model (, ,) do not improve the ATM fit, but they are not harmful either.

5.4 The β parameter prevents degeneracy in the log-moneyness model

In the displaced-diffusion model, even at ATM, different β values produce different call prices because the displacement shifts S and K by different amounts:

$$S' = S + \frac{\sigma_n(1-\beta)}{\sigma_{ln} \cdot \beta}, \quad K' = K + \frac{\sigma_n(1-\beta)}{\sigma_{ln} \cdot \beta}$$

When $\beta \rightarrow 1$, the state is almost log-normal; when $\beta \rightarrow 0$, it is almost normal. These different payoff profiles are distinguishable even at $S = K$, so the optimiser can meaningfully distribute weight across states.

5.5 Both models find that states 1 and 2 are “dead”

In the log-moneyness model, $\underline{\ln} = \underline{\ln} = 0.01$ (at the lower bound) across all regimes. The real dynamics live entirely in state 3. This is a broader finding: **the ATM backbone data only supports one active volatility state**, regardless of parameterisation.

Regime	Active State (Log-Moneyness)	Active State (Price-Level)
Low Vol	$w = 0.27, \underline{\ln} = 0.34$	$w = 0.61, \underline{N} = 400$
Transition	$w = 0.28, \underline{\ln} = 0.40$	$w = 1.0$
COVID	$w = 0.49, \underline{\ln} = 0.24$	$w = 1.0$
Post-COVID	$w = 0.29, \underline{\ln} = 0.52$	$w = 1.0$

5.6 Bottom line

Criterion	Log-Moneyness (DD Mixture)	Price-Level (Bachelier Mixture)
ATM fit quality	Same	Same
Parameter inter-pretability	High — captures skew/gear	Low — $\underline{N}$ values are arbitrary when w collapses
Multi-state identifiability at ATM	Yes (creates genuine separation)	No (degenerate: reduces to single state)
Parsimony	9 params (3 are s providing no ATM benefit)	6 params (3 are effectively unused)
OTM extension	Naturally handles skew via	Cannot generate skew — flat by construction
Verdict	Preferred — even at ATM, the s are not wasted; they enable the model to extend coherently to OTM strikes	

The pure Bachelier mixture is **not a viable alternative** for the uncertain volatility backbone. Its parameters are unidentifiable at ATM and it cannot produce skew away from ATM. The displaced-diffusion model’s $\underline{\ln}$ parameter — while irrelevant for the ATM fit itself — is essential for the model’s internal consistency and its ability to generalise to the volatility smile.